

Q3	
(a)	Gravitational potential at a point is the work done per unit mass in bringing a small test mass from infinity to that point (without a change in kinetic energy)
(b)(i)	To just reach the neutral point from the Earth, gain in gravitational potential energy GPE = Loss in kinetic energy $\Delta U = m\Delta\phi = -\Delta KE$ $10.0\Delta\phi = 6.10 \times 10^8 \text{ J}$ $\Delta\phi = 6.10 \times 10^7 \text{ J kg}^{-1}$ $\phi_{\text{neutral point}} - \phi_{\text{earth surface}} = 6.10 \times 10^7$ $\phi_{\text{neutral point}} = 6.10 \times 10^7 + \phi_{\text{earth surface}}$ $\phi_{\text{neutral point}} = 6.10 \times 10^7 + (-62.3 \times 10^6)$ $\phi_{\text{neutral point}} = -1.3 \times 10^6 \text{ J kg}^{-1} \text{ (2 s.f.)}$ OR total energy at Earth surface = total energy at neutral point $KE_{\text{earth surface}} + U_{\text{earth surface}} = KE_{\text{neutral point}} + U_{\text{neutral point}}$ $6.10 \times 10^8 + 10(-62.3 \times 10^6) = 0 - 10.0 \times \phi_{\text{neutral point}}$ $\phi_{\text{neutral point}} = -1.30 \times 10^6 \text{ J kg}^{-1} \text{ (3 s.f.)}$
(b)(ii)	The rock from the Moon must have enough energy to go past the neutral point, then resultant gravitational force of the Earth-mass system on the mass will accelerate it to the Earth. gravitational potential difference between the Moon's surface and the neutral point $\Delta\phi = \phi_{\text{neutral point}} - \phi_{\text{moon surface}}$ $= -1.30 \times 10^6 - (-3.90 \times 10^6)$ $= 2.60 \times 10^6 \text{ J kg}^{-1}$ Hence, the minimum kinetic energy needed to send a 1.4 kg rock from the Moon to the neutral point is $m\Delta\phi = (1.4)(2.60 \times 10^6) = 3.64 \times 10^6 \text{ J}$ or $3.6 \times 10^6 \text{ J}$ (2 s.f.)
(c)(i)	The gravitational force exerted on one star by the other star provides the centripetal force for each orbit. This pair of forces is an action-reaction pair (Newton's 3rd law), always equal in magnitude (and opposite in direction). OR The gravitational force between stars provides the centripetal force for each orbit. By Newton's Law of Gravitation, the gravitation force between the stars = $\frac{GM(2M)}{(3R)^2}$, thus the two stars experience the same magnitude of centripetal force.

(c)(ii)

$$\omega = \frac{2\pi}{T} = \frac{2\pi}{3.42 \times 10^5} = 1.837 \times 10^{-5} \text{ rad s}^{-1}$$

Consider the star of mass M ,

The gravitational force due to star of mass $2M$ provides the centripetal force for the orbit of M .

$$F_G = Ma_c$$

$$\frac{GM(2M)}{(3R)^2} = M(2R)\omega^2$$

$$R = \sqrt[3]{\frac{GM}{9\omega^2}}$$

$$= \sqrt[3]{\frac{6.67 \times 10^{-11} (3.14 \times 10^{30})}{9(1.837 \times 10^{-5})^2}}$$

$$= 4.10 \times 10^9 \text{ m}$$

